Kev v6 数据生成与续训实验报告

Qwen3 Qwen3.5

2026 10 30 v6

1 实验范围与主要结果

v6 Kev stage3 Qwen3 0.6B Base Qwen3.5 0.8B Base Kev Kev Qwen Kev [S1–S3]

v6 Qwen3 512/1000 509/1000 Qwen3.5 551/1000 585/1000 Qwen3.5

1 1 数据规模与配比

	100	100	6.67%	1.33%
	100	100	6.67%	1.33%
	350	720	23.33%	4.67%
	280	1400	18.67%	3.73%
	270	420	18.00%	3.60%
	300	600	20.00%	4.00%
	100	100	6.67%	1.33%
	1500	3440	100.00%	20.00%
	6000	7426	–	80.00%
	7500	10866	–	100.00%

: 1:4 1500 1500 1210
336 770 271 [S1 S4]

2 数据生成原则与保留类别

v6 Codex
LLM [S5]
hard-v1

2 1 三类原样保留

		v6
	31 25 24 20	100
	28 28 22 22	100
	24 22 22 JSON 18 14	100

300 90
28 12 through

2 2 概率题的生成与标注

350 75 60 40 50 25 40 60 720
2–5 0% 100% 1 10 100 1000 10000

“ ” “ ” 50%

1% 80% 5% 0.01×0.80 ÷ (0.01×0.80
0.99×0.05) 13.91% [10%, 30%) “ ” ID train/bayes/25

3 信息不足与多跳数据

3 1 信息不足通过多个问法监督同一决定

v5 280 40 20 16
4 140 112 28 140 280

	Yes No Cannot be determined

5 1400 840 560

30 2027-11-30 2028-01-23
54 “ ” v5 insufficient/train/000007

3 2 多跳补充不同关系规则

	90	180
	60	120
	30	30
	45	45
	45	45
	270	420

180 90 v5 2 3 4
50%

Violet Violet→Sienna→Hazel Hazel Sienna
ID train/exact_supervisor_hops/00010

4 长政策数据与质量检查

4 1 长政策保留情境并修订候选答案

v5 300 150 600 120 120 60 180 120 subtype v6 JSON “ ”

$2316 $2000 $75 $1925 subtype not_purchased v5 policy/train/000471

4 2 标签与数据隔离检查

	0
	0
4923	0
	3140
	680
	7500 10866

facts [S1 S4 S5]

v6

5 模型与续训方法

Kev stage3 LoRA adapter pointer head v5 v6 stage4 LoRA 256 pointer head Transformers PyTorch Kev RL vLLM [S2 S6]

	Qwen3-0.6B-Base Qwen3.5-0.8B-Base
	stage3 adapter head
LoRA	rank 16 targets=all placement=full
	Qwen3 10.6M Qwen3.5 11.3M
	AdamW LR 1e-5 weight decay 0.01
Head	head_lr=0 LR 1e-5
	OneCycleLR pct_start=0.1 1.0
	1 epoch seed 101
	batch 4 × accum 2 8
	938 4
	BF16 weights_dtype=fp32
	max_state 7552 branch 8192 packed 9216
	perm_kl ord_w anchor_w Brier focal 0

5 1 数据回放与损失计算

decision-v7 12576 random.Random("replay:101") 6000 1500 6000

choice noul score score ord_w=0 1400/3440

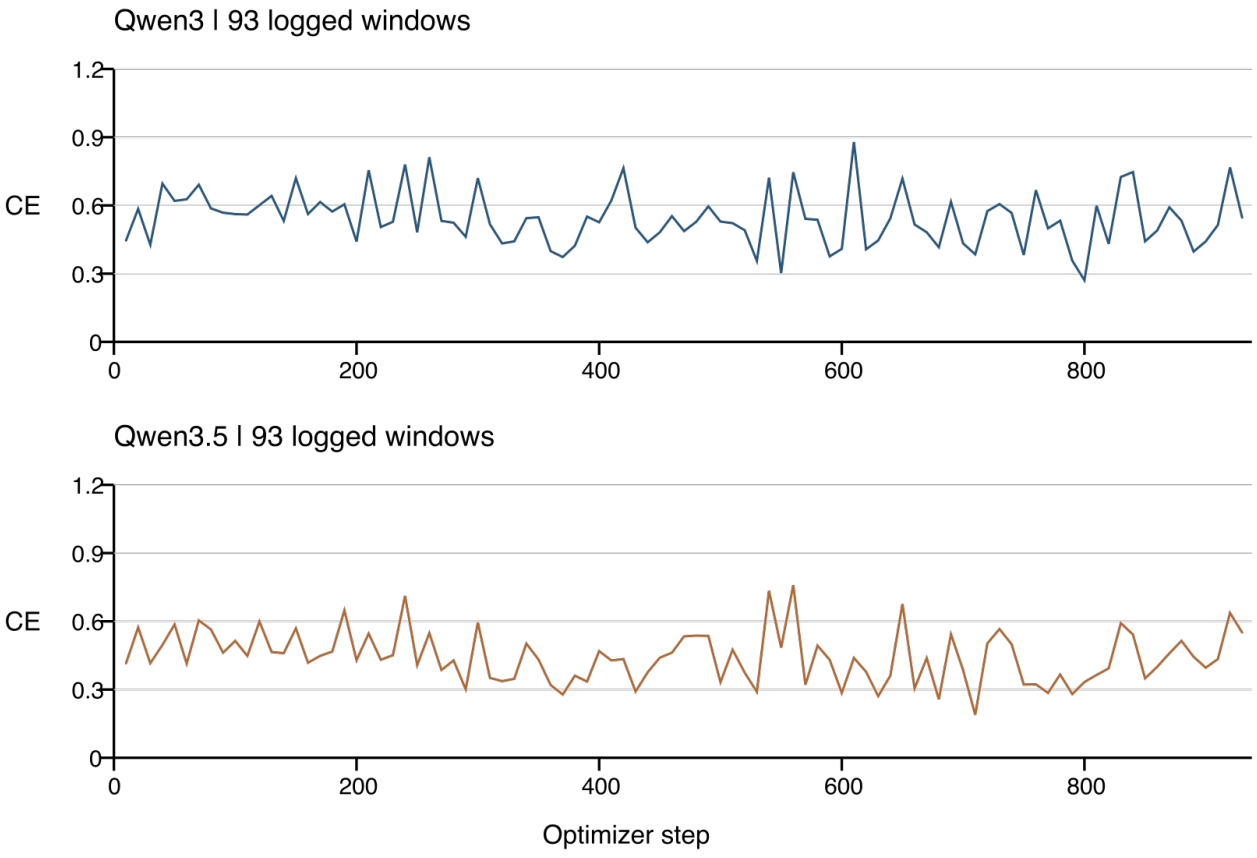
5 2 训练增强

p_none=0.10 p_none_distract=0.12 p_distract=0.15 p_none_pair=0.25 synthetic_repeat=1 none 45 none one-hot target none 9498 7500

6 □□□□□□□□□

	Qwen3	Qwen3.5
	7,500	7,500
	9,498	9,498
	938	938
token	2,707,325	2,719,099
	14 29	16 58
	8.23 GiB	10.75 GiB
	0 0	0 0

A800 938 wall_seconds 868.914 1017.575
nvidia-smi



1 v6

10 93 step 10–930 938 8 loss
pipeline [S6]

7 真实测试题上的前后对比

hard-v1 devtools-v1 800 1000 argmax
pp “ ” stage3 [S3]

		Qwen3 → %	pp	Qwen3.5 → %	pp
	150	34.67→36.00	+1.33	22.67→34.67	+12.00
	117	37.61→41.88	+4.27	43.59→41.88	-1.71
	158	53.80→53.80	+0.00	55.06→53.16	-1.90
	100	66.00→61.00	-5.00	69.00→70.00	+1.00
	125	48.80→44.00	-4.80	68.00→66.40	-1.60
	200	58.50→58.00	-0.50	65.50→67.50	+2.00
	150	58.00→59.33	+1.33	62.67→74.67	+12.00
	1000	51.20→50.90	−0.30	55.10→58.50	+3.40

7 1 原错题修复和新增错误

Qwen3	488	43	8.81%	46	-3
Qwen3.5	449	57	12.69%	23	+34

Qwen3 43 46 3 Qwen3.5 57 23 34 Qwen3.5
18 2

7 2 完整测试集结果

			→ %	pp
hard-v1	Qwen3	1088	57.17→54.69	-2.48
hard-v1	Qwen3.5	1088	64.98→64.43	-0.55
devtools-v1	Qwen3	1073	63.00→63.47	+0.47
devtools-v1	Qwen3.5	1073	65.14→68.97	+3.82

hard-v1 Qwen3.5 devtools-v1 Kev
devtools ID

8 哪些能力改善以及哪些仍未学会

8 1 四类重点问题的细分结果

		Qwen3	Qwen3.5
	39	36→38	34→38
	61	30→23	35→32
	26	13→8	17→13
	25	8→7	7→10
	35	16→14	17→16
	35	8→9	8→10
	65	65→65	65→65
	11	2→2	1→1

Qwen3

90 45

Qwen3.5

[S7]

8 2 合成训练探针与开发集

	Qwen3	Qwen3.5	Qwen3	Qwen3.5
	13→16/38	15→17/38	38→47/124	32→55/124
	48→58/100	42→67/100	170→213/350	197→261/350
	8→8/30	10→15/30	32→33/86	41→44/86
	15→18/40	15→26/40	33→34/120	35→61/120

20 140 268 “ ” Qwen3 8/30 “ ” Qwen3.5 [S7]

Qwen3 309/770 378/770 40.13%→49.09% Qwen3.5 341/770 480/770 44.29%→62.34%

9 已完成对照与结论边界

v6 stage3 [S8]

		Qwen3	Qwen3.5
stage3		512/1000	551/1000
v5	1500	514/1000	584/1000
v6		509/1000	585/1000
v6	v6 epoch 3	516/1000	582/1000
v6	140	504/1000	584/1000

v5 v6 Qwen3 5 Qwen3.5 1 Qwen3 7 Qwen3.5 3

9.1 可以支持的结论

v6 Qwen3.5 3.4 Qwen3

9.2 不能支持的结论

Kev seed 101

95% bootstrap Qwen3 −2.09 +1.58 pp Qwen3.5 +1.65 +5.23 pp 14

9.3 对后续决策的含义

stage3

10 □□□□□□□□□

manifest.json status run_status

procogs/artifacts/kev_training_eval_20261026/targeted_v6_20261030
procogs/tools /C20545/kev-08b-official-repro-20261026

10 1 □□□□

S1 manifest.json train.jsonl development.jsonl train_probe.jsonl frozen_input_hashes.json

S2 evidence/runs/qwen3-20261030-targeted-v6/stage4/ qwen35-20261030-targeted-v6/stage4/ training_config.json training_metrics.json

S3 comparison-seven.json comparison-seven.csv paired-seven.jsonl bootstrap

S4 preflight_qwen3.json preflight_qwen35.json none_guard_records.json none_augmentation_with_replay.json

S5 tools/build_targeted_v6.py build_kev_probability_coverage_candidate.py build_kev_insufficient_multiview_candidates.py build_kev_multihop_coverage_candidates.py build_kev_policy_fixed_options_candidate.py

S6 evidence/stages/stage3/kev/train.py evidence/targeted_v6_20261030/pipeline_qwen3.log pipeline_qwen35.log loss

S7 all_seven_mechanism_diagnostics.json train_probe_comparison.json development_pair_diagnostics.json

S8 relative_to_v5_guard.json targeted_v6_3ep_20261030/ targeted_v6_factmix_candidate_20261030/ comparison-seven.json

10 2 □□□□□□□

train.jsonl SHA256
`58572b2118b10d9505d5bdb027585fab75f4ad7e52a1c5d8d155979459fecea8`

development.jsonl SHA256
`5747c5b02565f10688d270d8da5afa384386e9cfc42bf55e2971d2924810fe63`

v6-verified-evidence.tar.gz SHA256
`4e833233af5596a368c55e11b720210c950f876b7a2c3950ae3b01e36ab1b980`

84 runs/< >-20261030-targeted-v6/stage4
JSONL SHA256